

A Civic Stethoscope for **HEALTHY CITY**

Utilizing 311 Requests to Monitor Neighborhood Health Determinants



Baoqi Eileen Chen

Department of City & Regional Planning,
College of Environmental Design,
University of California, Berkeley



Junhong Duan

Spatial Sciences Institute,
Dornsife College of Letters, Arts and Sciences,
University of Southern California

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Propose — an interdisciplinary argument

Present — a literature review

Ask for — your thoughts and feedback



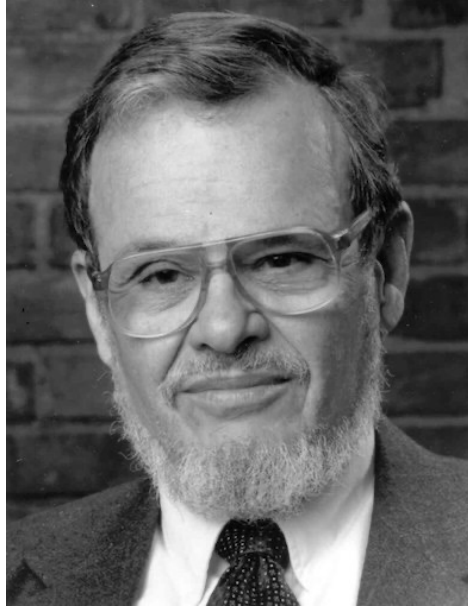
- 1 Motivation** | Why this? Why now?
- 2 Healthy City** | Leonard J. Duhl's vision
- 3 Civic Stethoscope** | What is it? How do we use it?
- 4 Preliminary Results** | What's been done? By us, and by all of us
- 5 Next Steps** | Where to head from here?

1 Motivation

*A healthy city is greater than the sum of its parts,
the urban and the urbanites.*

1 Motivation

Leonard J. Duhl



Health and the city

by Professor Len Duhl

School of Public Health, University of California, Berkeley, USA

(Duhl 1990)

1 Motivation

HEALTH PROMOTION
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Vol 1, No. 1
Printed in Great Britain

The healthy city: Its function and its future

LEONARD J. DUHL

*Professor of Public Health and City Planning, University of California, Berkeley and
Clinical Professor of Psychiatry, University of California, San Francisco*

SUMMARY

The author describes urban health and what constitutes a healthy city. Emphasis is placed on the importance of the various parts of the city/organism communicating with each other and remaining responsive and flexible. The healthy city/organism must be competent for each of its members/parts and for the community as a whole.

Several overall requirements that are presented as important for a healthy city are: that the city's

responses to its developmental needs, its organizations and its people be appropriate and effective; that the city have the ability to modify itself and change to meet the always emerging, changing requirements for life; that it be competent so that its inhabitants use it; and that it perform an educational role for its citizens.

In presenting actions to create a more healthy city the relative value of responding to the symptoms and dealing with underlying issues to create a context for success is discussed, as are specific actions.

(Duhl 1986)

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communicating

healthy city/organism

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modify itself

always emerging

responding to the symptoms

*responses to its developmental needs, its organizations and its people be appropriate and effective; that the city have the ability to **modify itself** and change to meet the **always emerging**, changing requirements for life; that it be competent so that its inhabitants use it; and that it perform an educational role for its citizens.*

*In presenting actions to create a more healthy city the relative value of **responding to the symptoms** and dealing with underlying issues to create a context for success is discussed, as are specific actions.*

(Duhl 1986)

1 Motivation



emerging spatial data

revisit



*Unprecedented
empirical precision*



1 Motivation

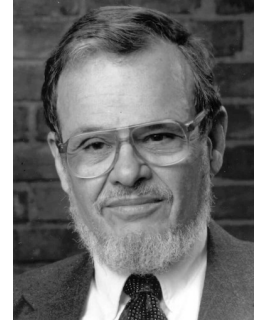


emerging spatial data

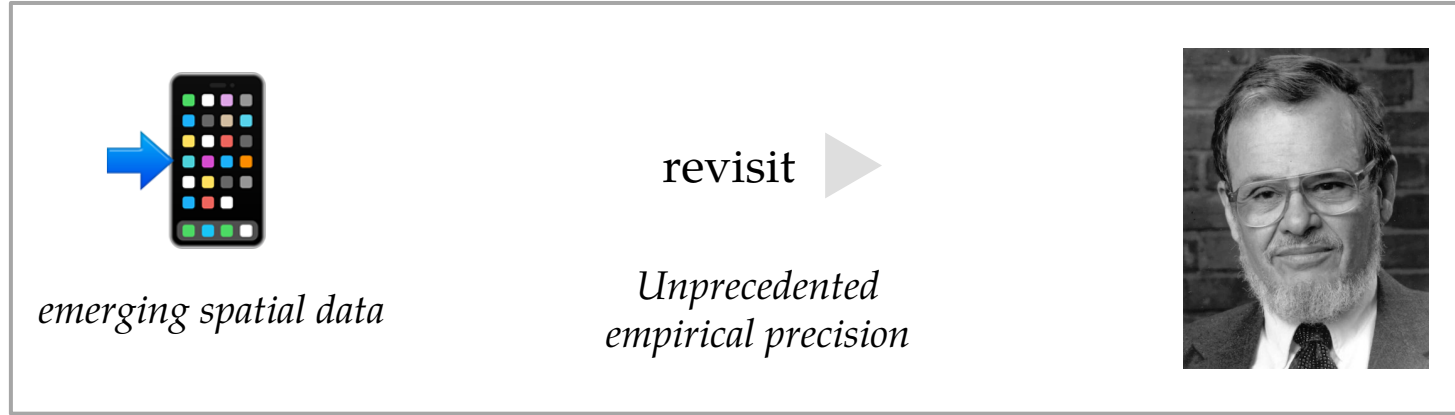
revisit



*Unprecedented
empirical precision*



1 Motivation



Need for new instruments



*theoretical
methodological*

1 Motivation



2 Healthy City

REQUIREMENTS FOR A HEALTHY CITY

- 1 — Ability to provide the developmental needs
- 2 — Ability to cope with breakdown, to change and modify
- 3 — Competence leading to ability of individuals or groups to use it
- 4 — Ability to perform its educational role

ACTIONS TO CREATE A MORE HEALTHY CITY

- 1 — Respond to the symptoms
- 2 — Deal with the underlying issues
- 3 — Specific actions

(Duhl 1986)

2 Healthy City



Complex/Dynamical
System Perspective

Community/Place-Based

Reciprocal Management
Urban Governance

3 Civic Stethoscope

The “311”

a U.S. municipal service for citizens to access information, report non-urgent issues, and request assistance through phone calls, mobile apps, and online platforms

Austin, TX
Baltimore, MD
Baton Rouge, LA
Berkeley, CA
Birmingham, AL
Boston, MA
Buffalo, NY
Chicago, IL
Detroit, MI
Evanston, IL
Kansas City, MO
Little Rock, AR
Los Angeles, CA
Louisville, KY
Memphis, TN

Minneapolis, MN
Montgomery, AL
Nashville, TN
New Orleans, LA
New York City, NY
Oakland, CA
Philadelphia, PA
Pittsburgh, PA
Riverside, CA
Sacramento, CA
San Francisco, CA
Somerville, MA
Tacoma, WA
Washington, D.C.

- | | |
|-------------------------|-------------------------------|
| 1. Abandoned Vehicles | 16. Parks |
| 2. Animals-Dead | 17. Potholes |
| 3. Animals-Live | 18. Rodents/Pests |
| 4. Building Complaints | 19. Sewage |
| 5. Bulk items | 20. Signs |
| 6. Code enforcement | 21. Snow/Ice Removal |
| 7. Dumping | 22. Stormwater Drainage |
| 8. Flooding | 23. Street & Sidewalk Repairs |
| 9. Garbage/Recycling | 24. Street Cleaning |
| 10. Graffiti | 25. Streetlights |
| 11. Hazardous Materials | 26. Taxes/Payments |
| 12. Homelessness | 27. Traffic Signals |
| 13. Mowing | 28. Transit |
| 14. Noise | 29. Trees |
| 15. Parking | 30. Water |

(N.L. Stowers 2022)

3 Civic Stethoscope



Public Accessibility

By-street-and-minute
Resolution

Decade-long,
National Coverage

4 Preliminary Results

PRISMA Analysis

Database

Web of Science
PubMed
Scoups
Science Direct

Keywords

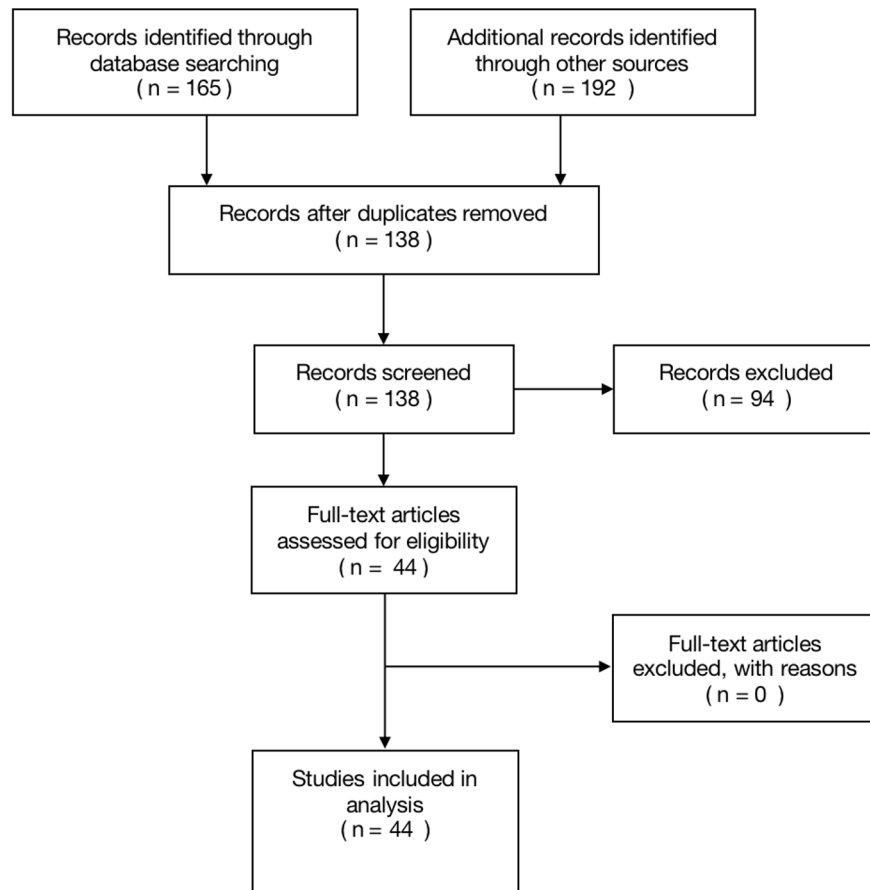
Title, abstract, keywords: "311 non-emergency" OR "311 Calls" OR "311 system" OR "non-emergency service" OR "311 service" OR "constituent relationship management" OR "311 Complains" OR "311 request" OR "311 request service"

Identification

Screening

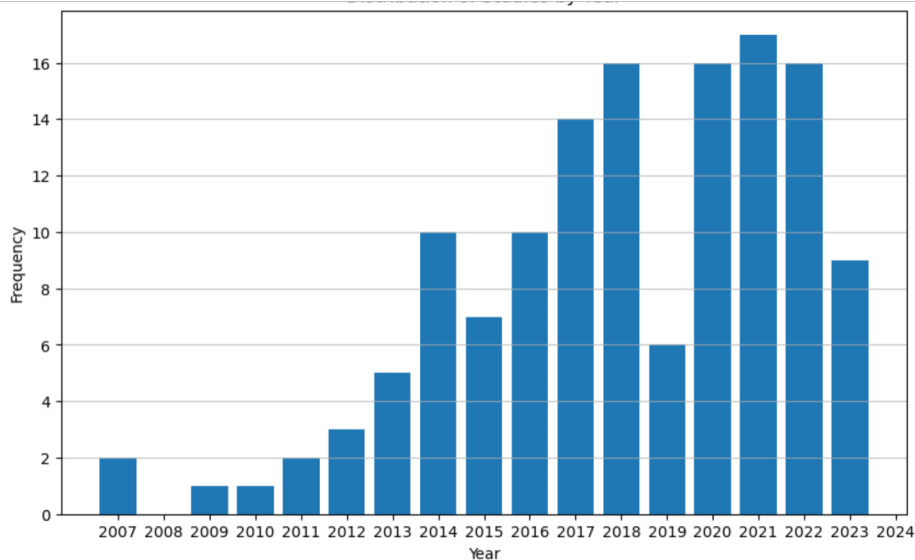
Eligibility

Included

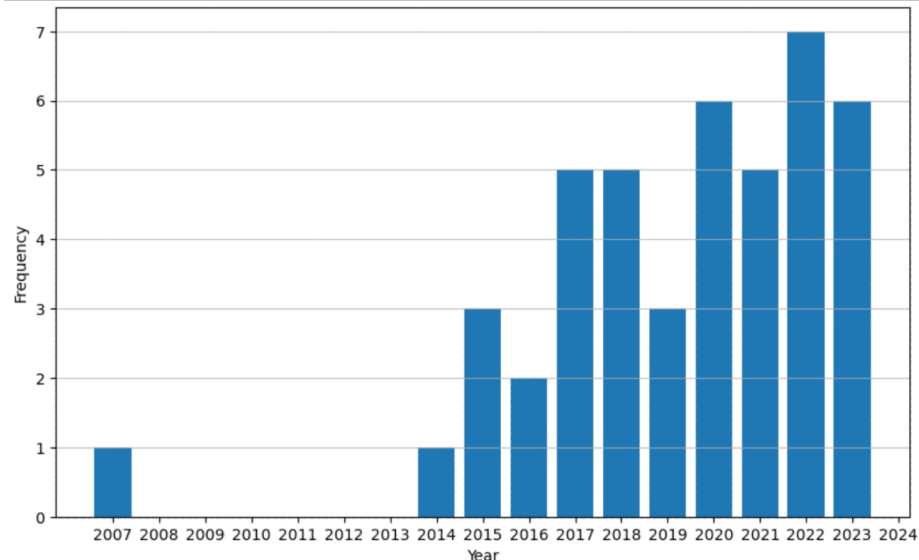


4 Preliminary Results

Overall Trend



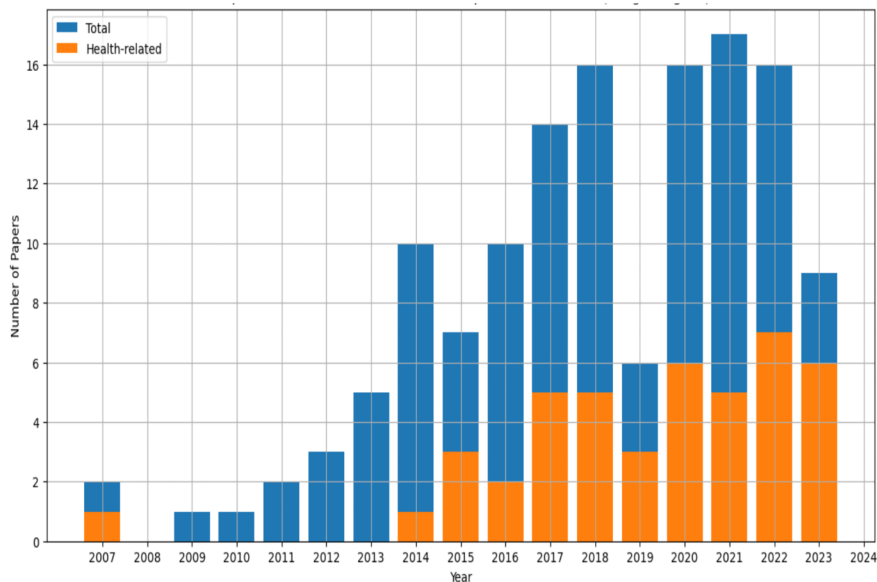
Total 311 Studies By Year



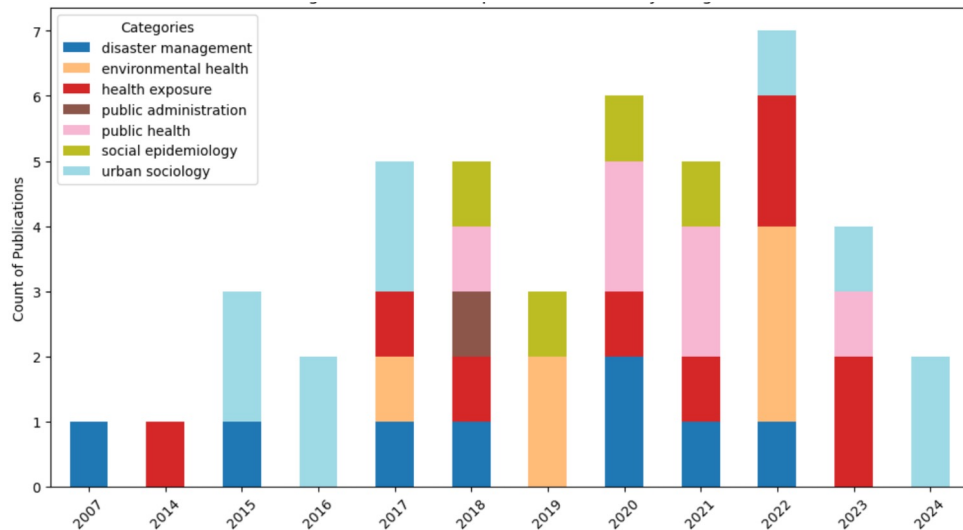
311 Studies on Health By Year

4 Preliminary Results

A Zoom-in Look



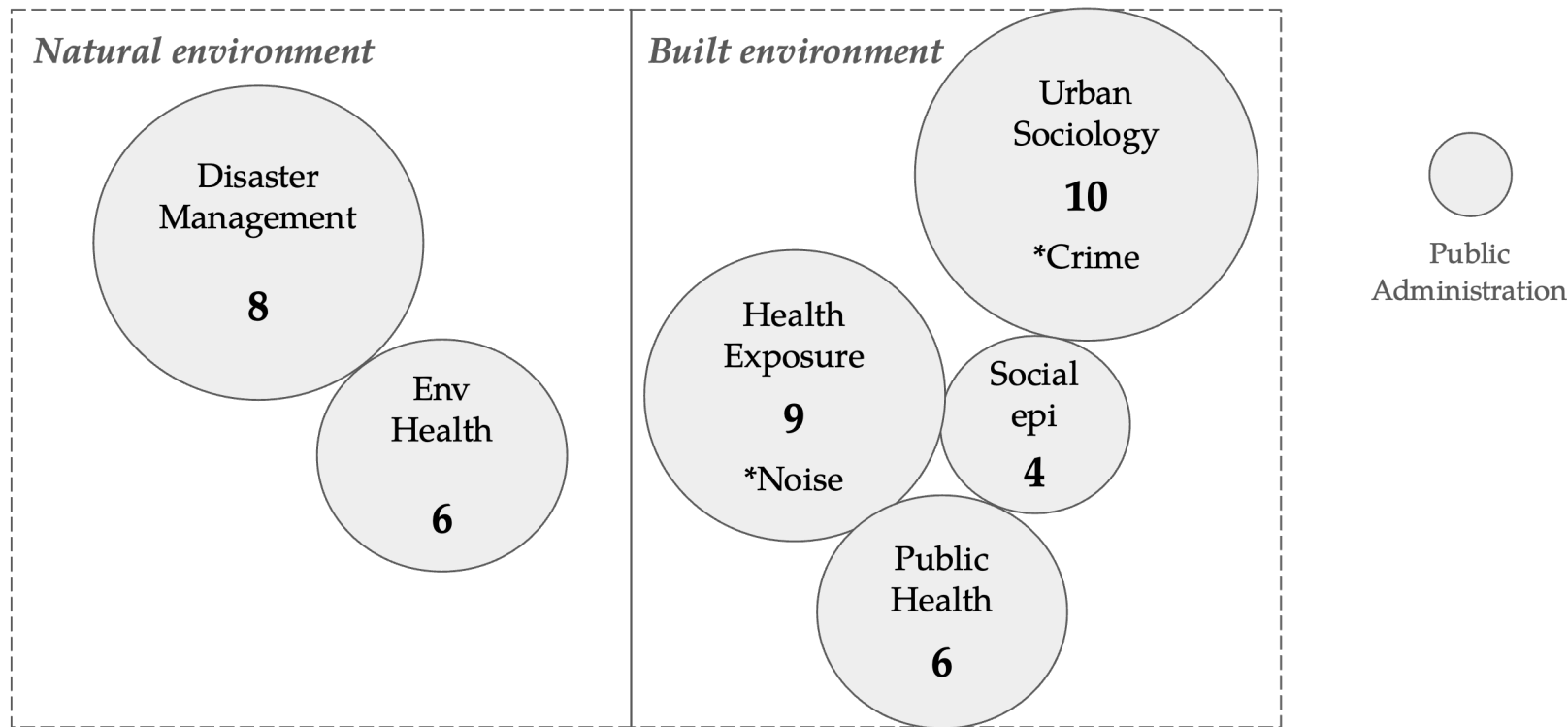
Proportion of 311 Studies on Health By Year



311 Studies on Health Method Breakdown By Year

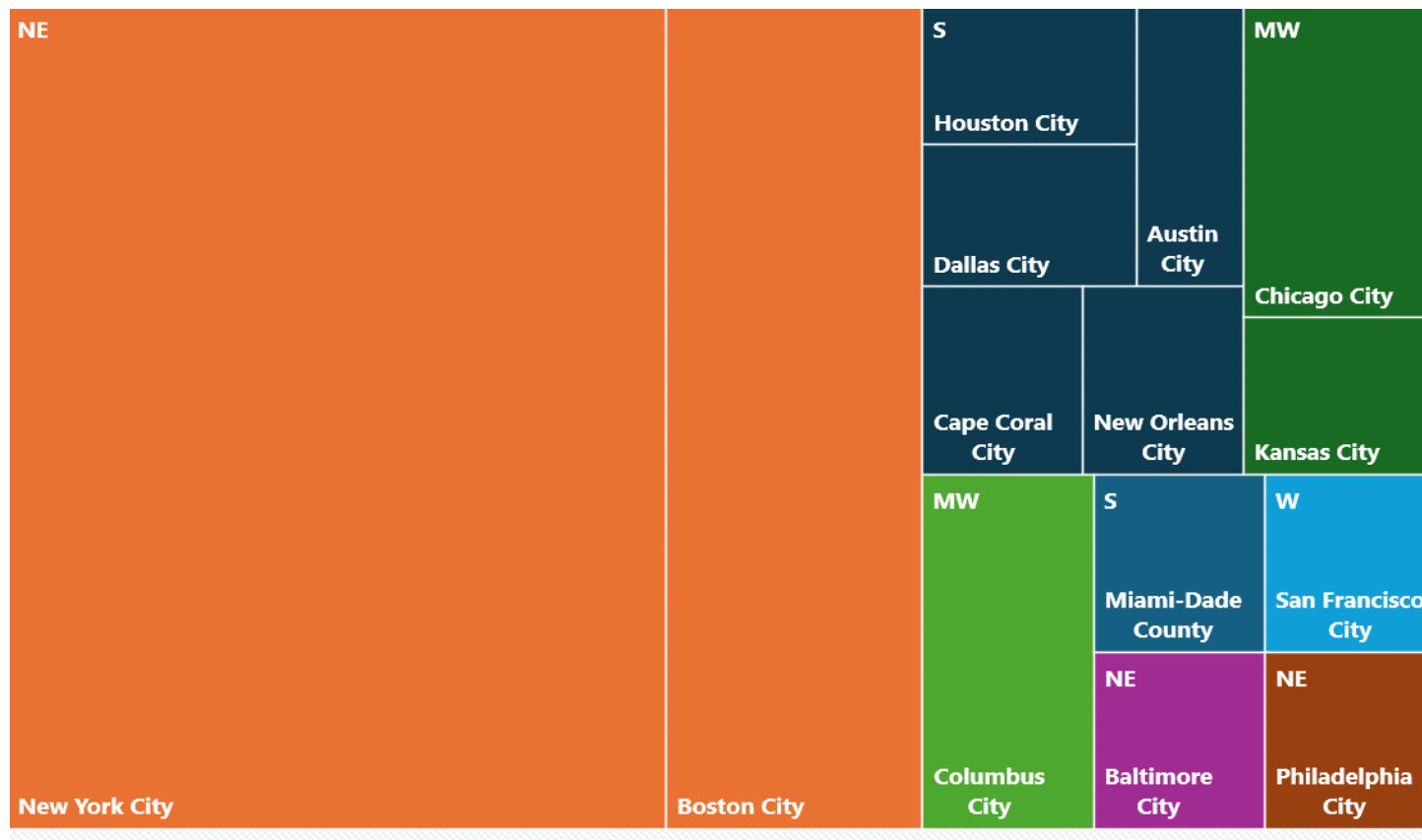
4 Preliminary Results

Discipline/Topic Communities



4 Preliminary Results

By Region



4 Preliminary Results

Word Cloud of Request Categories



4 Preliminary Results

Spatial Unit of Analysis

Resolution
Low



High

Administrative

City-level = 3

Zip Code = 4

Census tract
= 12

Census block groups
= 4

Spatial-statistical

Hexagon = 1

Thiessen = 1

Local

Region = 1

Boroughs = 1

FSA = 1

Street seg = 1

Tax lot = 1

4 Preliminary Results

Temporal Unit of Analysis

Resolution
Low



High

Yearly = 9

Monthly = 12

Weekly = 7

hourly = 5

daily = 13

4 Preliminary Results



Research Discipline

Methodology for
Request Categories

Spatiotemporal
Resolution Selection

5 Next Steps

- ▶ Current research connects 311 data to specific issues like crime and noise; future studies could explore a broader range of health outcomes.
- ▶ Many studies focus on specific types of data or isolated incidents. A more holistic integration of various data types, including 311 data with other municipal and social data sources, could provide deeper insights.
- ▶ A need for longitudinal studies that track the impact of neighborhood conditions reported through 311 over time on community health
- ▶ Expanding research to include diverse urban settings can enhance the generalizability of findings.

Email: baoqiche@berkeley.edu | **X:** @BaoqiEileenChen | **LinkedIn:** Baoqi Eileen Chen
Email: junhongd@usc.edu | **X:** @Mardy_Duan | **LinkedIn:** Junhong (Mardy) Duan

Thank you!



Baoqi Eileen Chen

Department of City & Regional Planning,
College of Environmental Design,
University of California, Berkeley

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Spatial Sciences Institute,
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References

Duhl, Leonard J. 1986. "The Healthy City: Its Function and Its Future." *Health Promotion International* 1 (1): 55–60. <https://doi.org/10.1093/heapro/1.1.55>.

Duhl, Leonard J. 1990. "Health and the City / by Len Duhl." <https://policycommons.net/artifacts/609682/health-and-the-city-by-len-duhl/1589280/>.

N.L. Stowers, Genie. 2022. "Back to Basics: City Services and 311 Service Requests." *State and Local Government Review* 54 (1): 13–31. <https://doi.org/10.1177/0160323X211064253>.

*** full list of reviewed literature in a separate sheet*